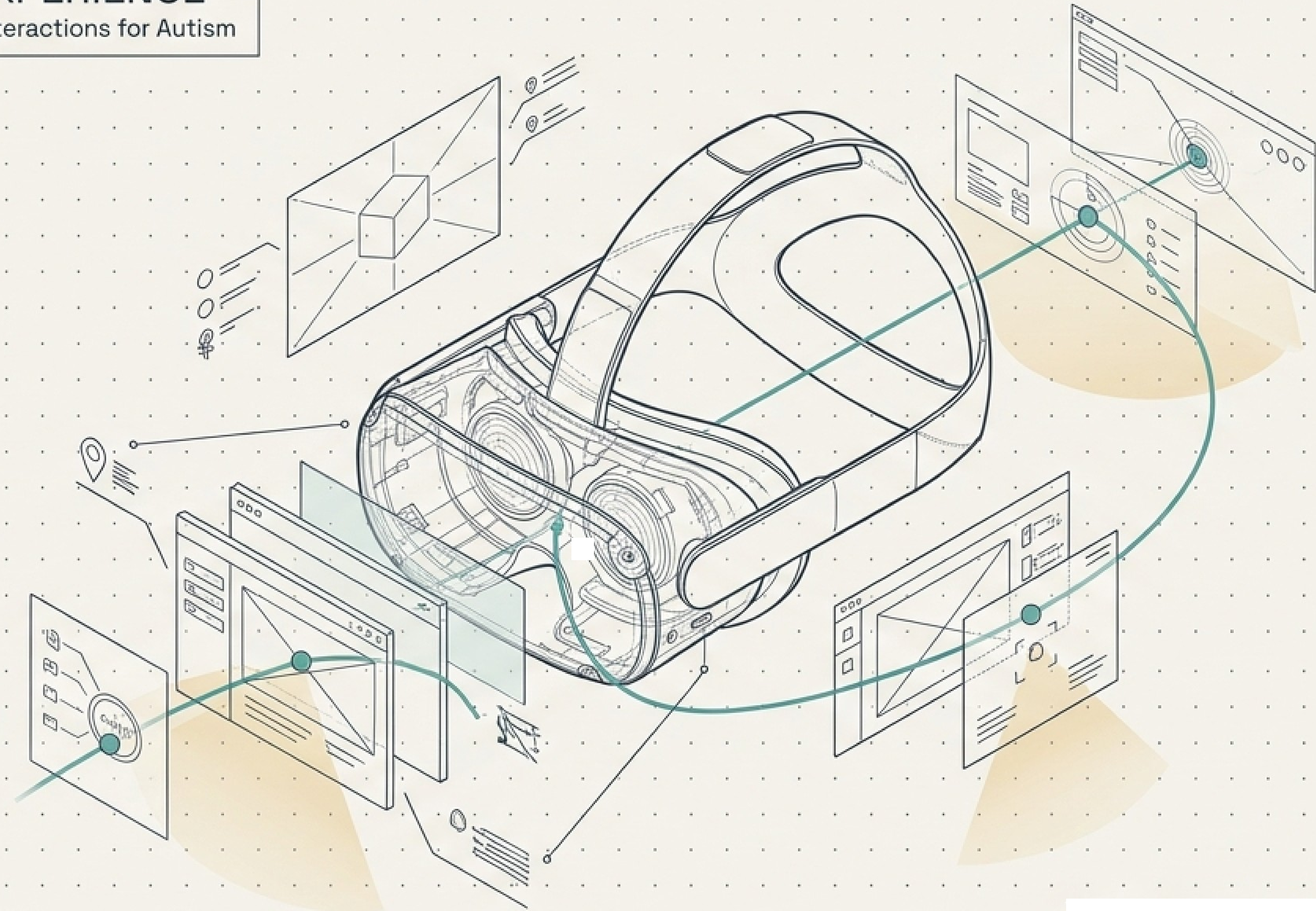


INCLUSIVITY: A VR EXPERIENCE

Designing Sensory-Safe Spatial Interactions for Autism



Standard virtual reality assumes a neurotypical sensory baseline. This project engineers a new spatial paradigm for users on the autism spectrum.

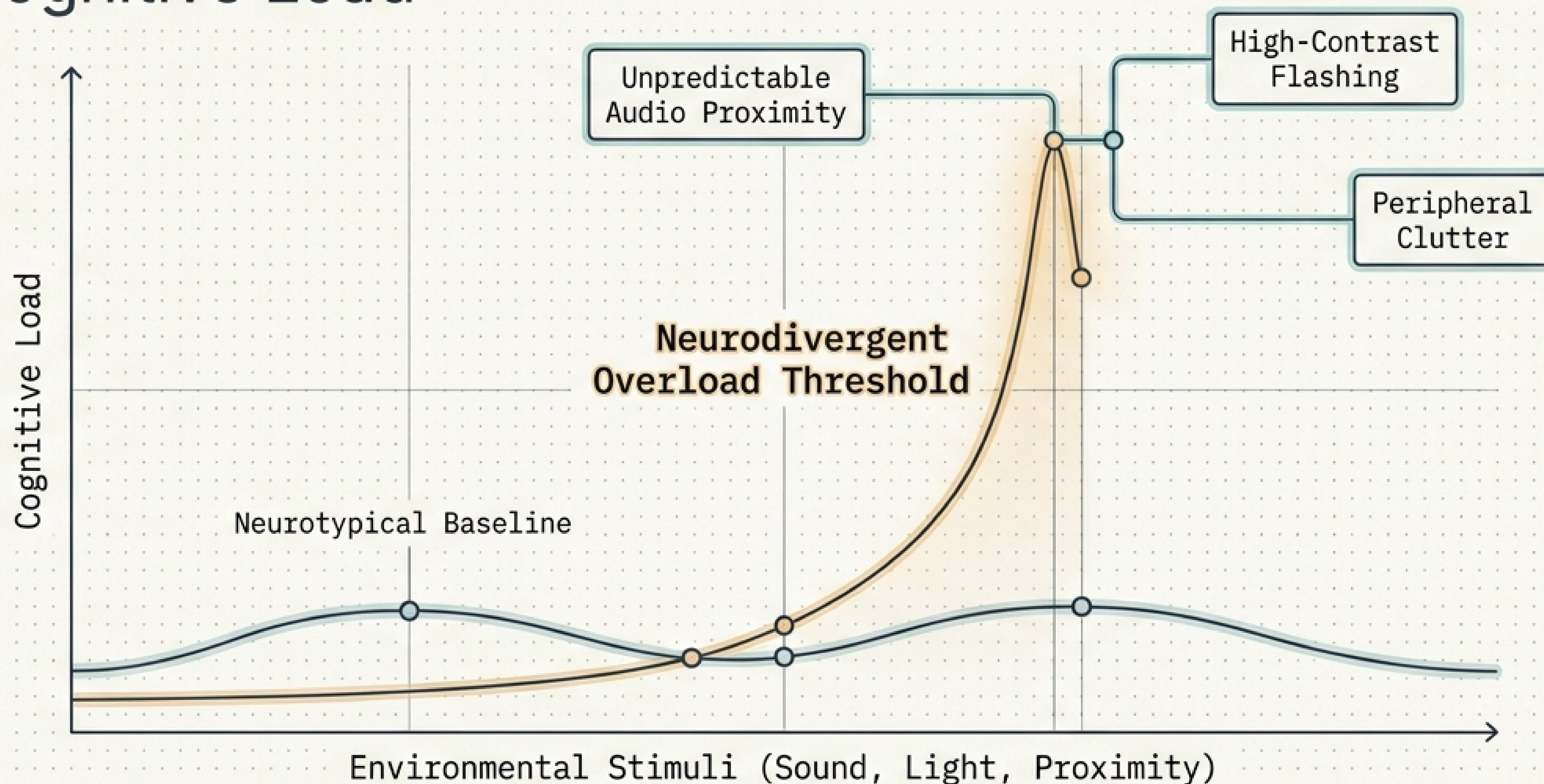
[Interaction Design]

[Oculus]

[Blender 3D]

[Figma]

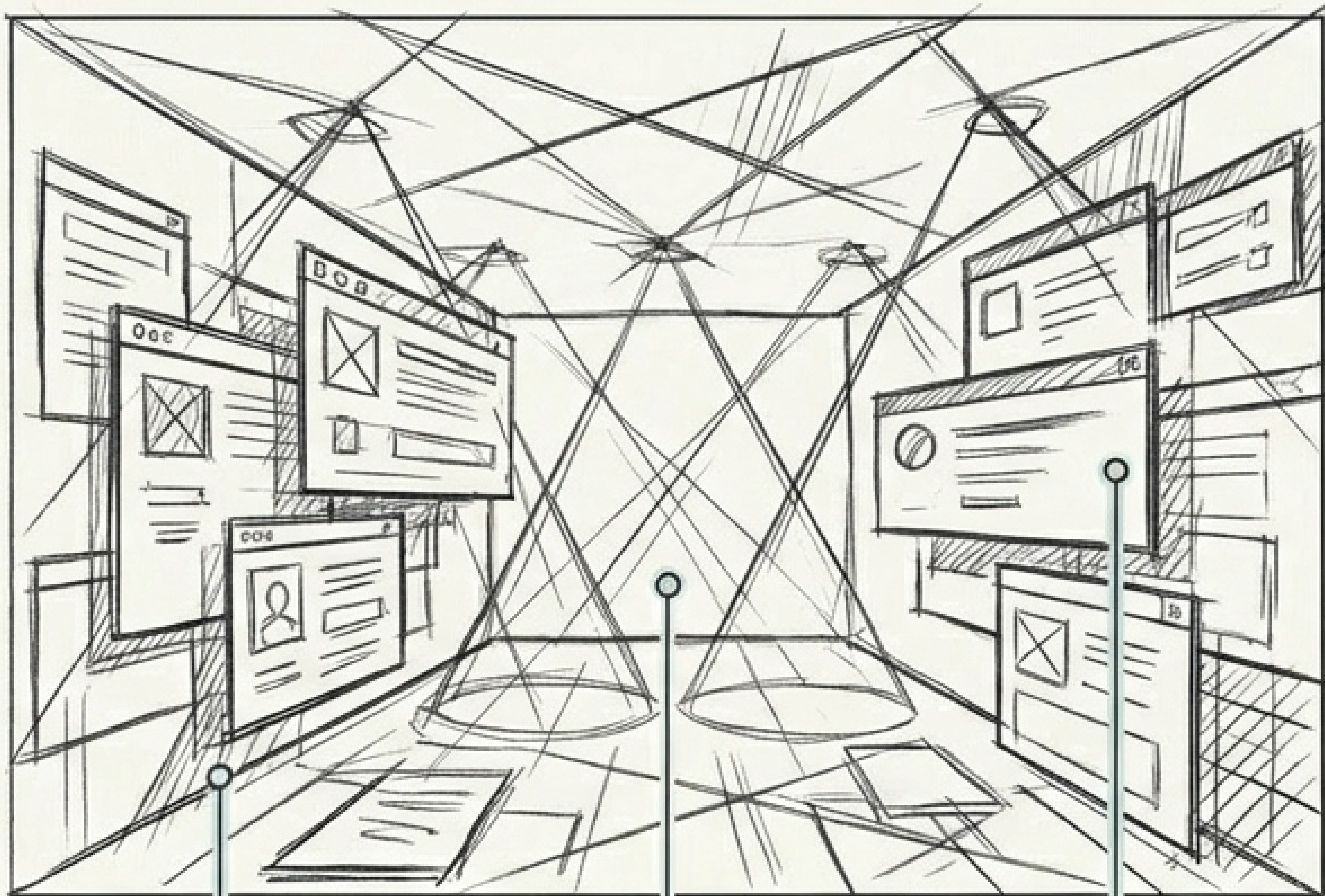
Mapping the Sensory Spectrum and Cognitive Load



Traditional spatial environments trigger rapid sensory overload, requiring a fundamental architectural shift in XR.

Standard Immersion vs. Inclusive Spatial Architecture

STANDARD VR



Lighting

Harsh &
High-Contrast

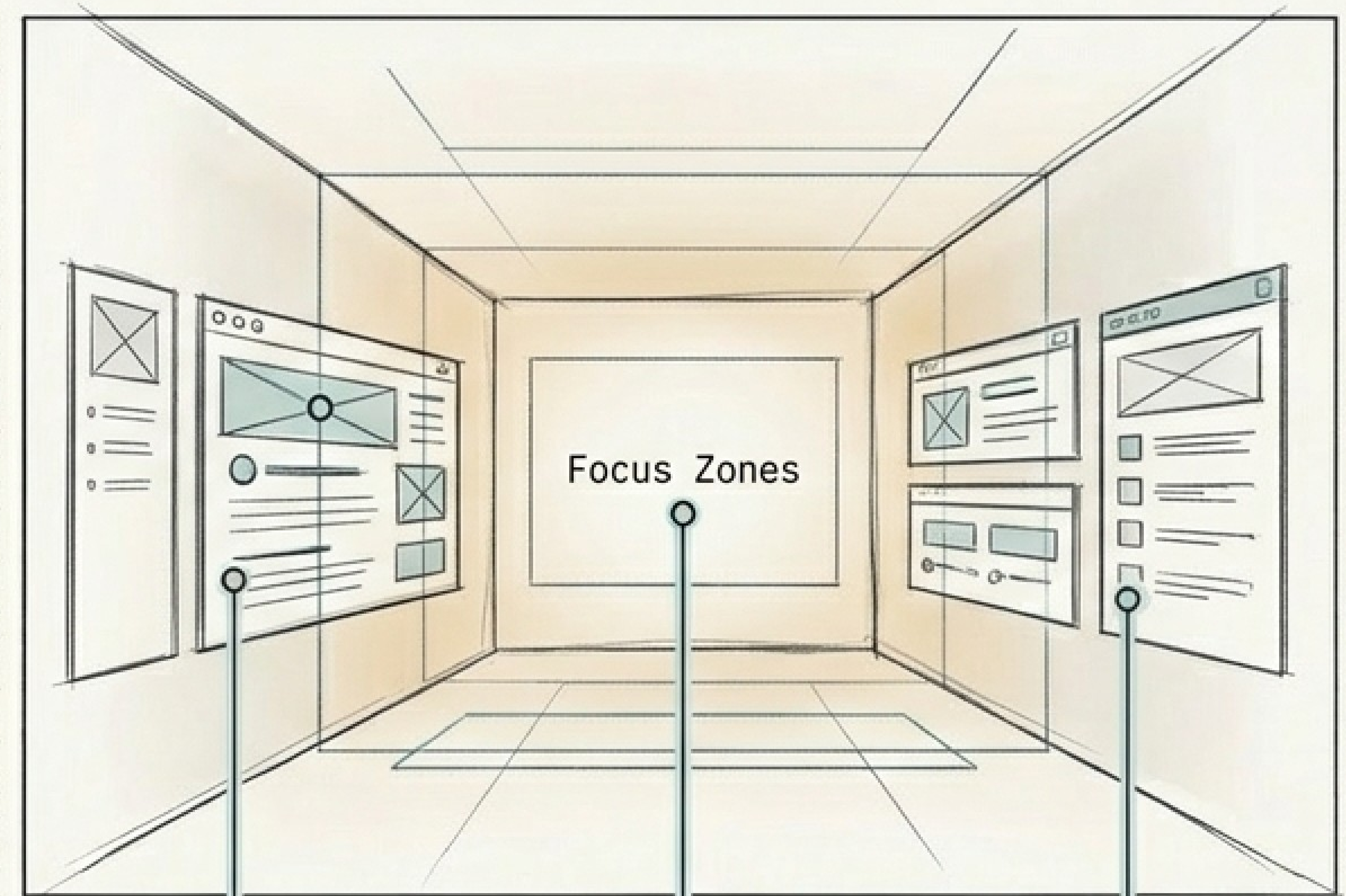
Audio

Omnidirectional
Overlap

UI Anchoring

Intrusive &
Unpredictable

INCLUSIVE VR



Lighting

Soft Ambient
Gradients

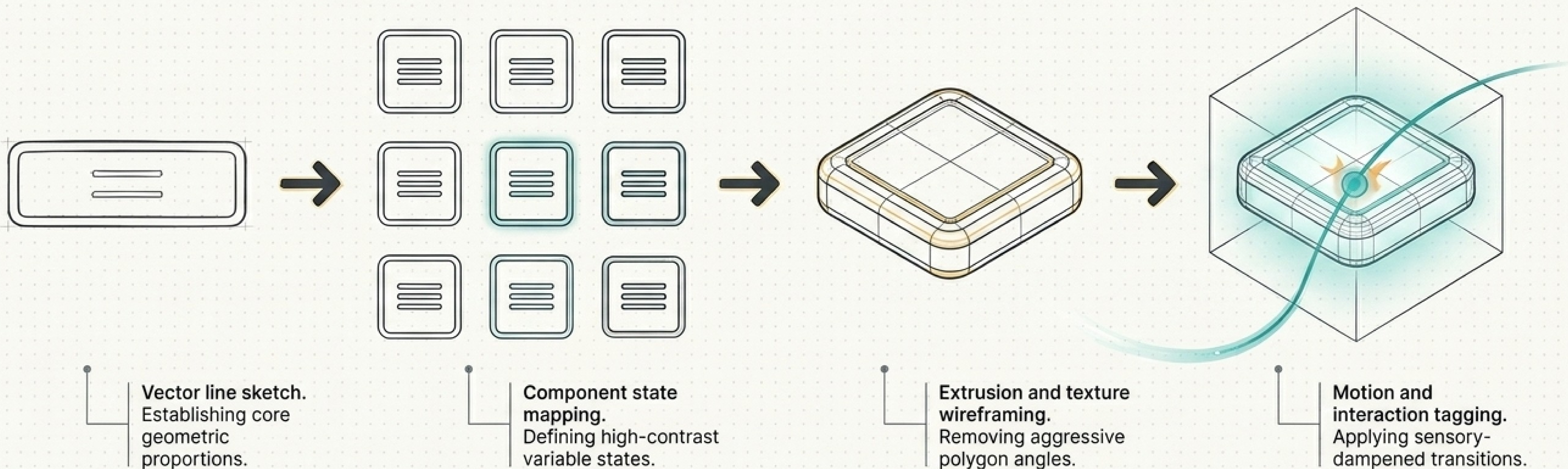
Audio

Point-Dampened
Isolation

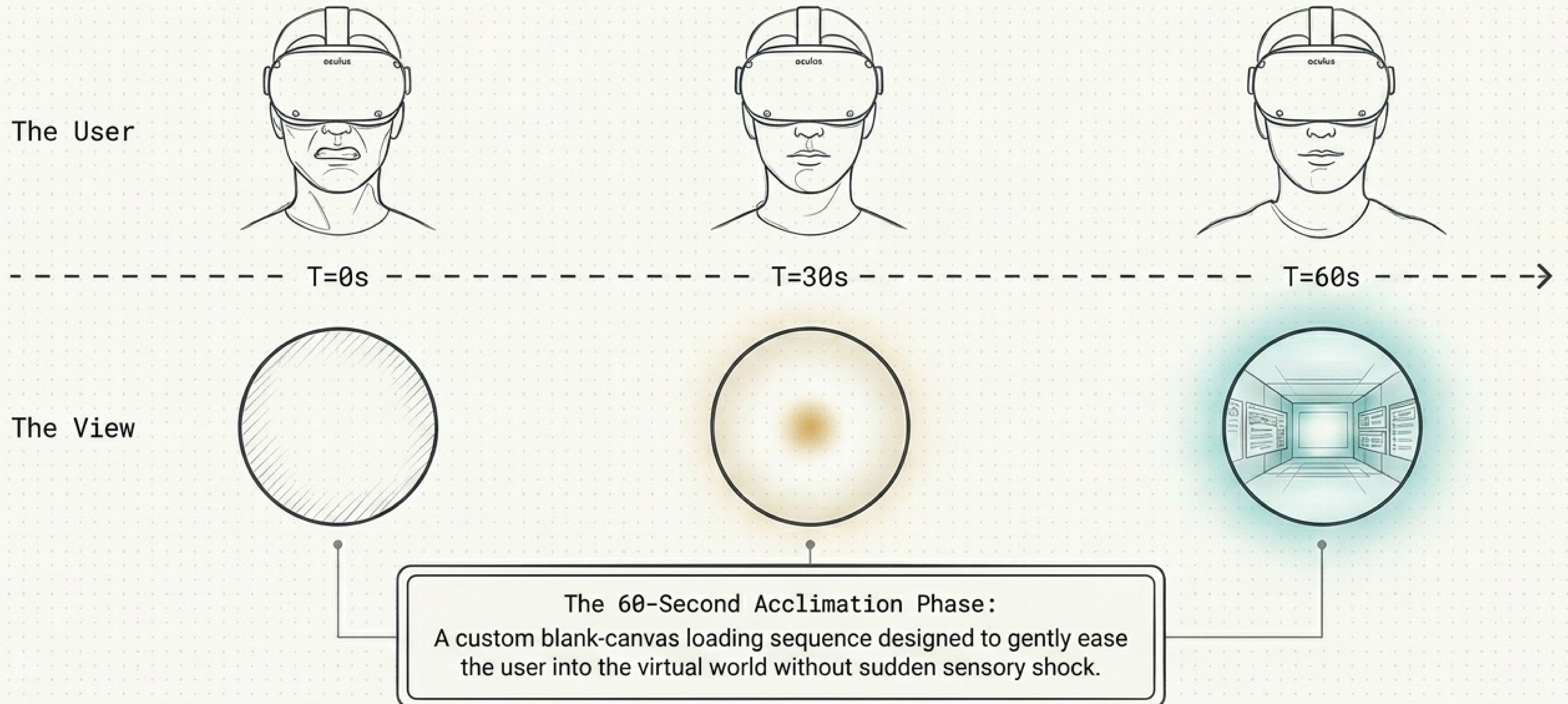
UI Anchoring

Predictable
Bounds

The 2D-to-Spatial Asset Pipeline

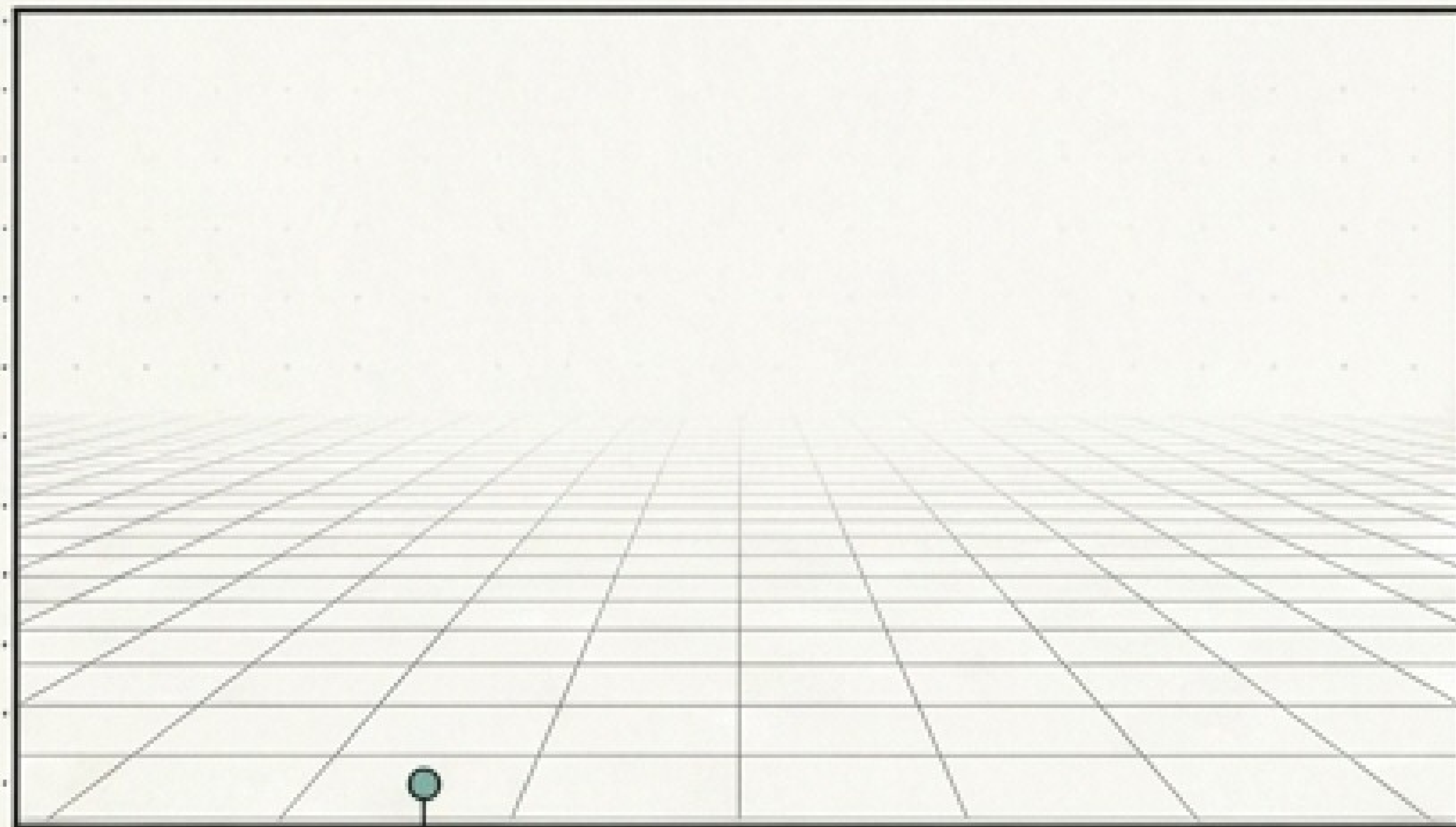


The Acclimation Phase and User Onboarding



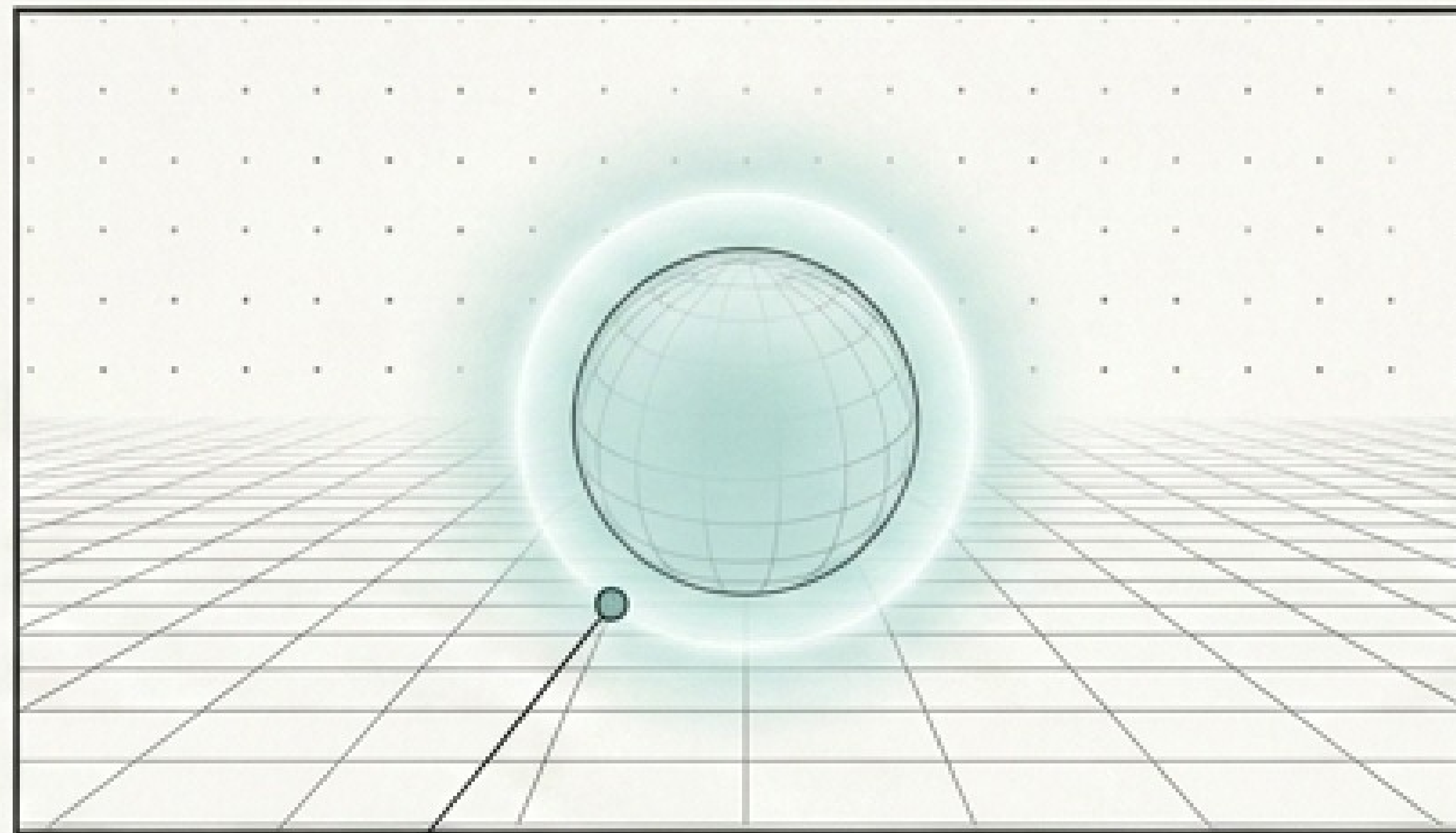
Entering the Space Gradually

A



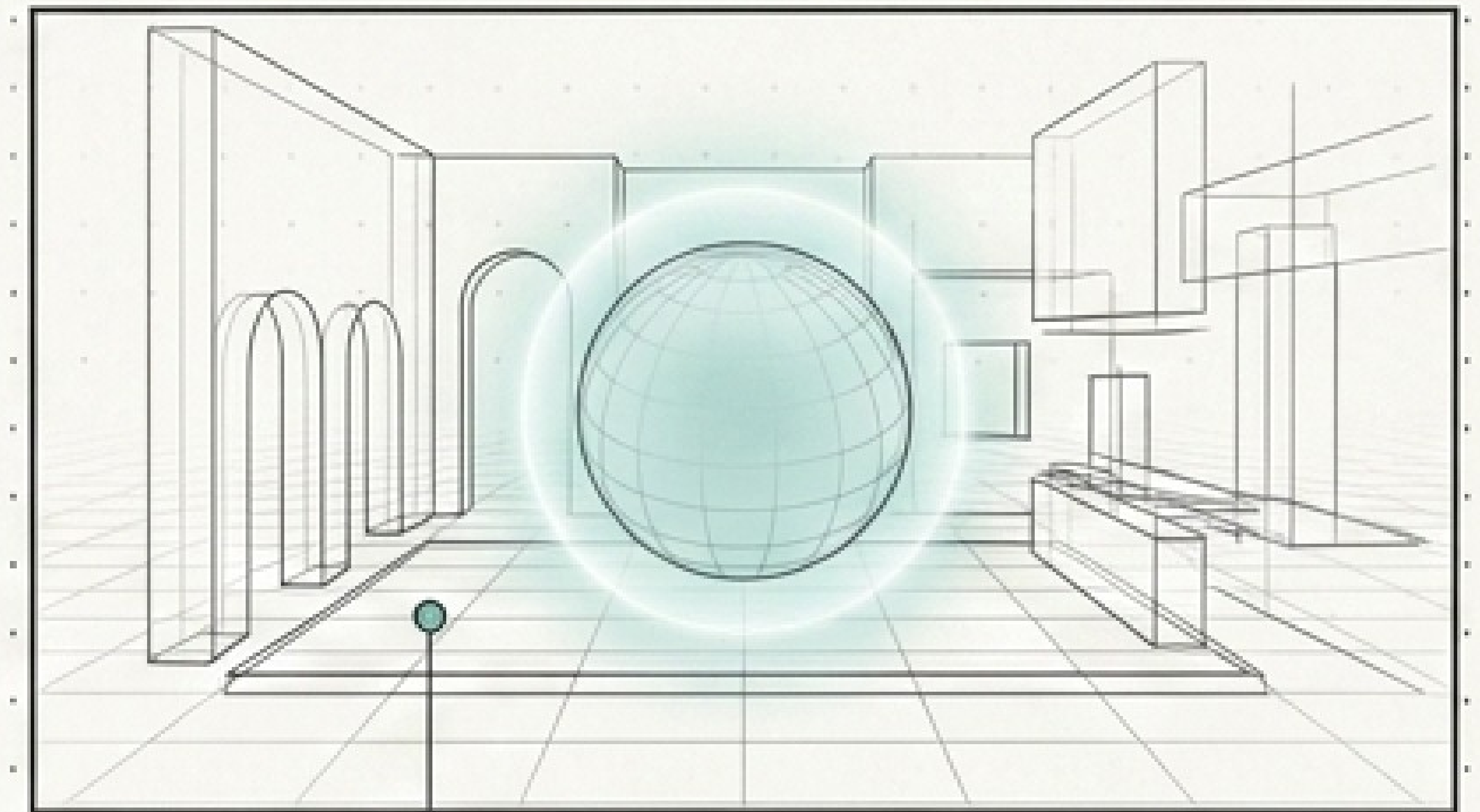
Zero-gravity easing

B



Restricted
color gamut

C

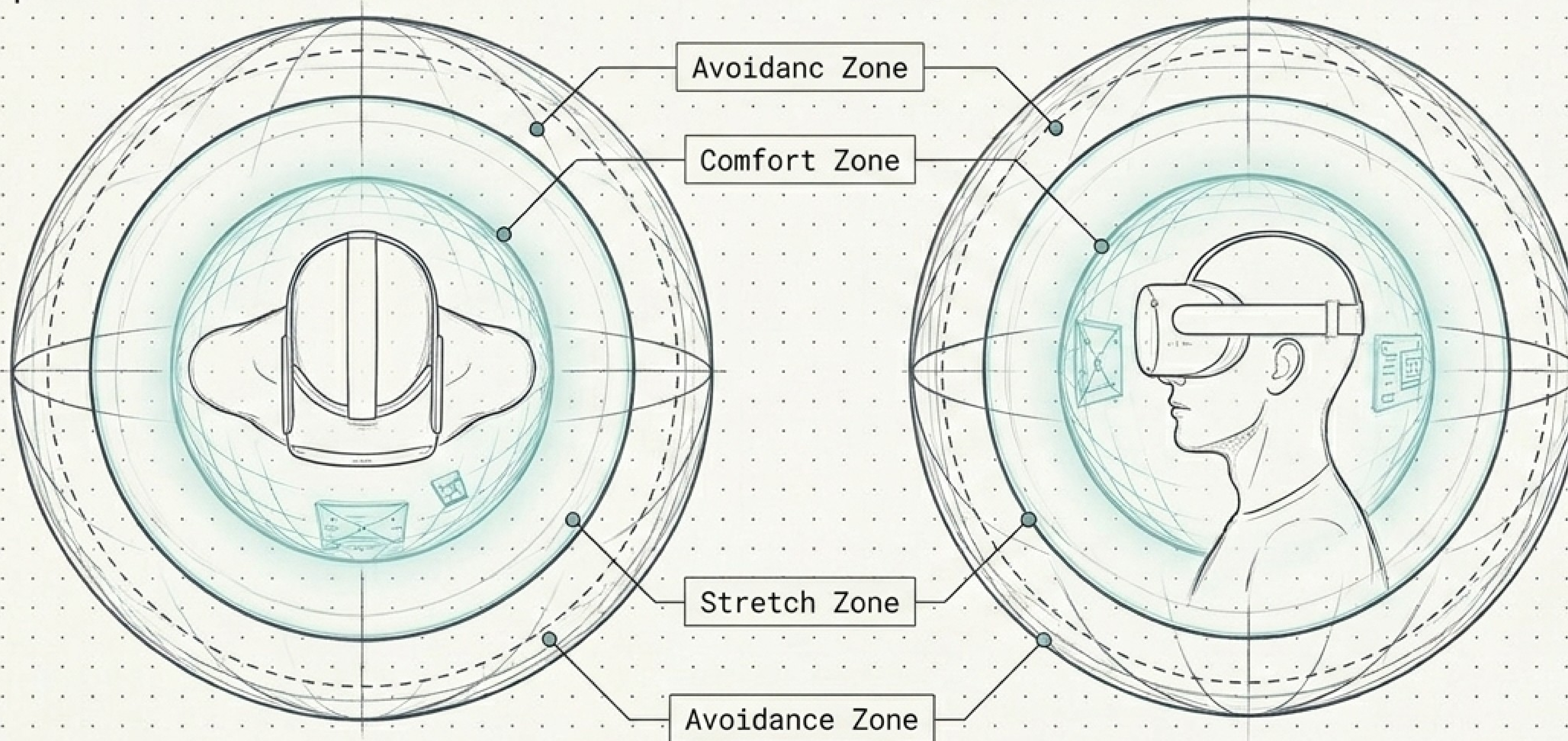


Gaze-triggered
rendering progression

Spatial Ergonomics within the Reach and Gaze Model

Top-Down View

Side-Profile View



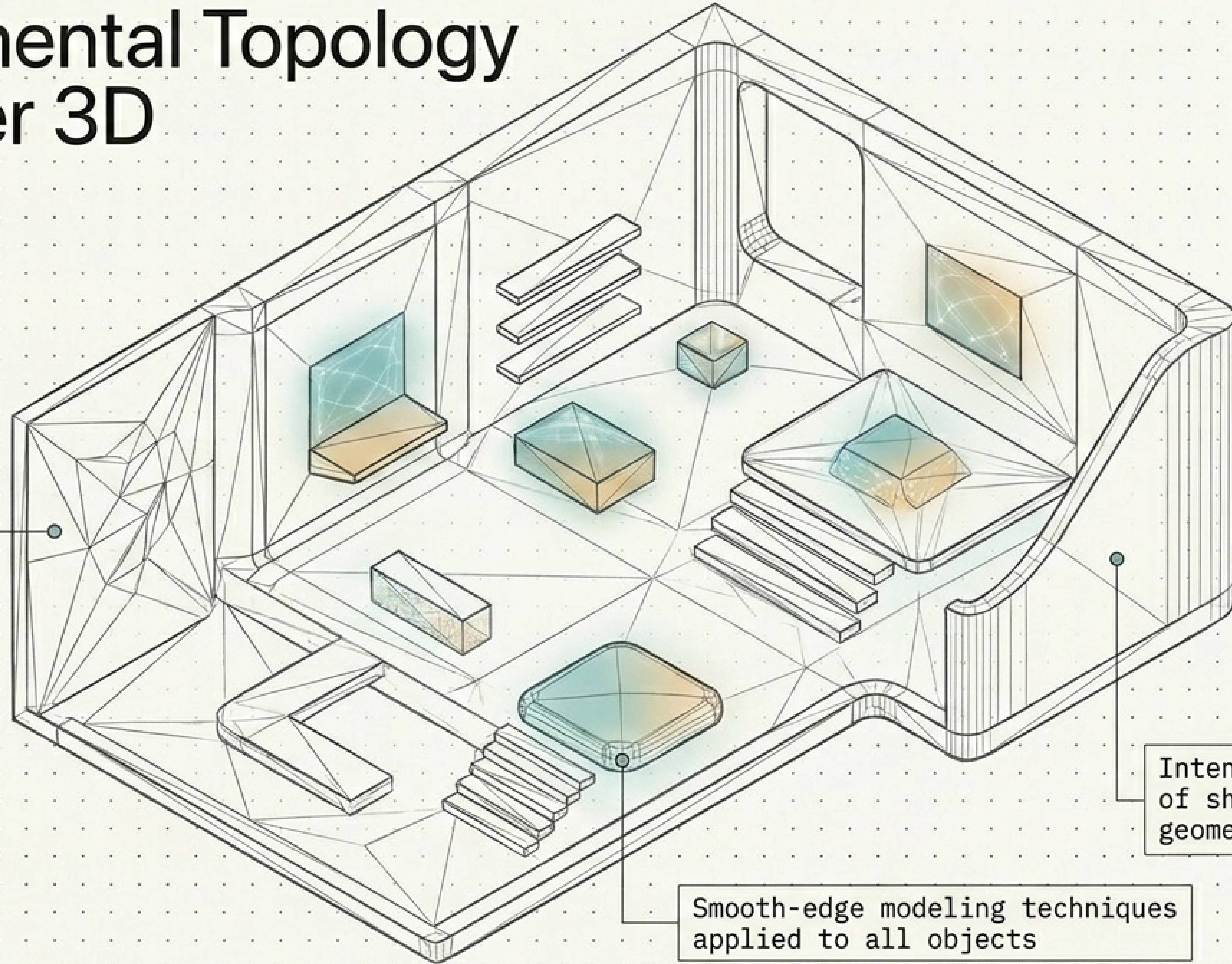
Neurodivergent motor-spatial preferences require strict boundaries. All interactive UI panels are rigidly constrained to the 'Comfort Zone' to prevent physical anxiety, fine-motor fatigue, and spatial disorientation.

Environmental Topology in Blender 3D

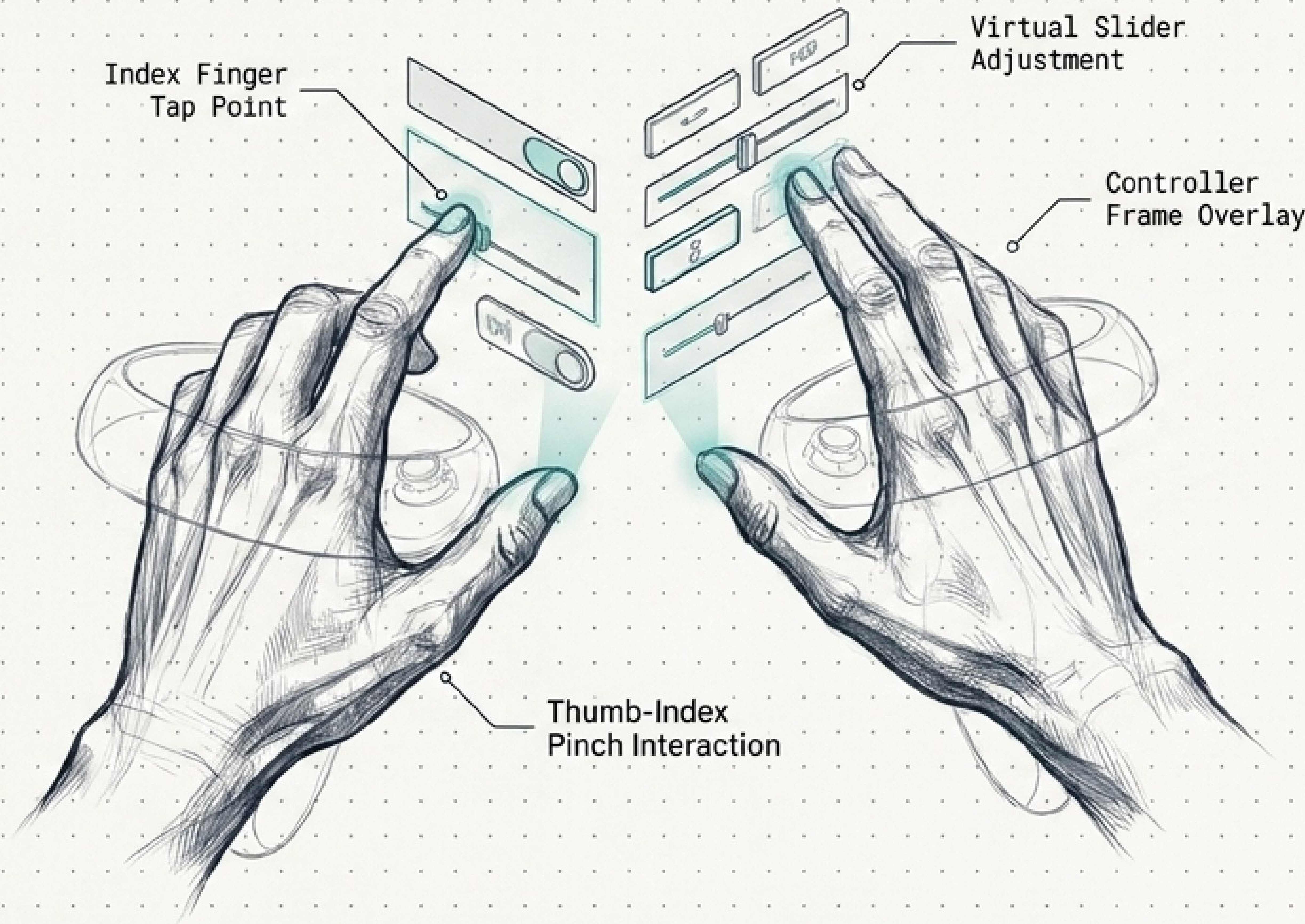
Strict polygon
limits

Intentional avoidance
of sharp, aggressive
geometric angles

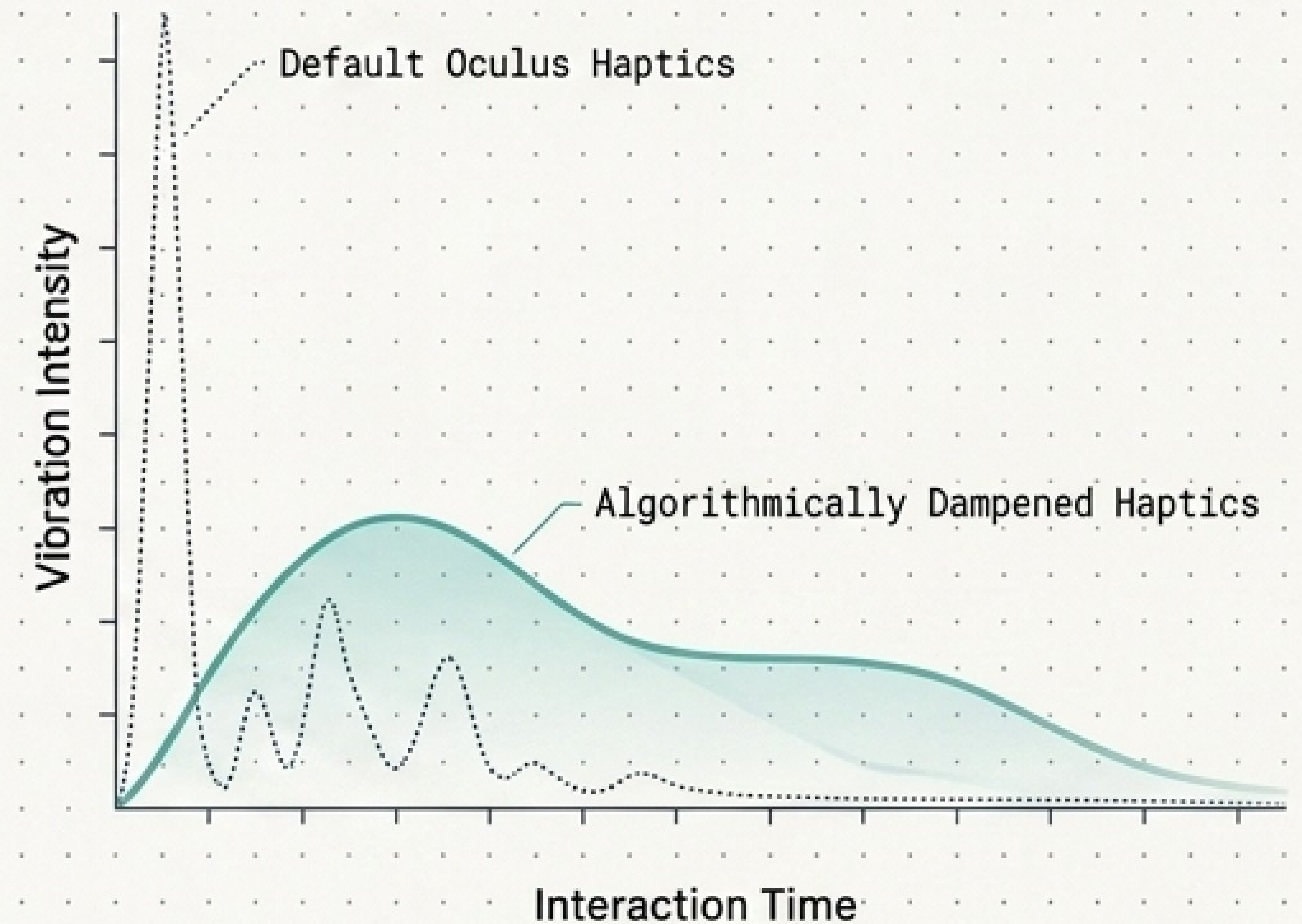
Smooth-edge modeling techniques
applied to all objects



Hand Tracking, Micro-Interactions, and Dampened Haptics

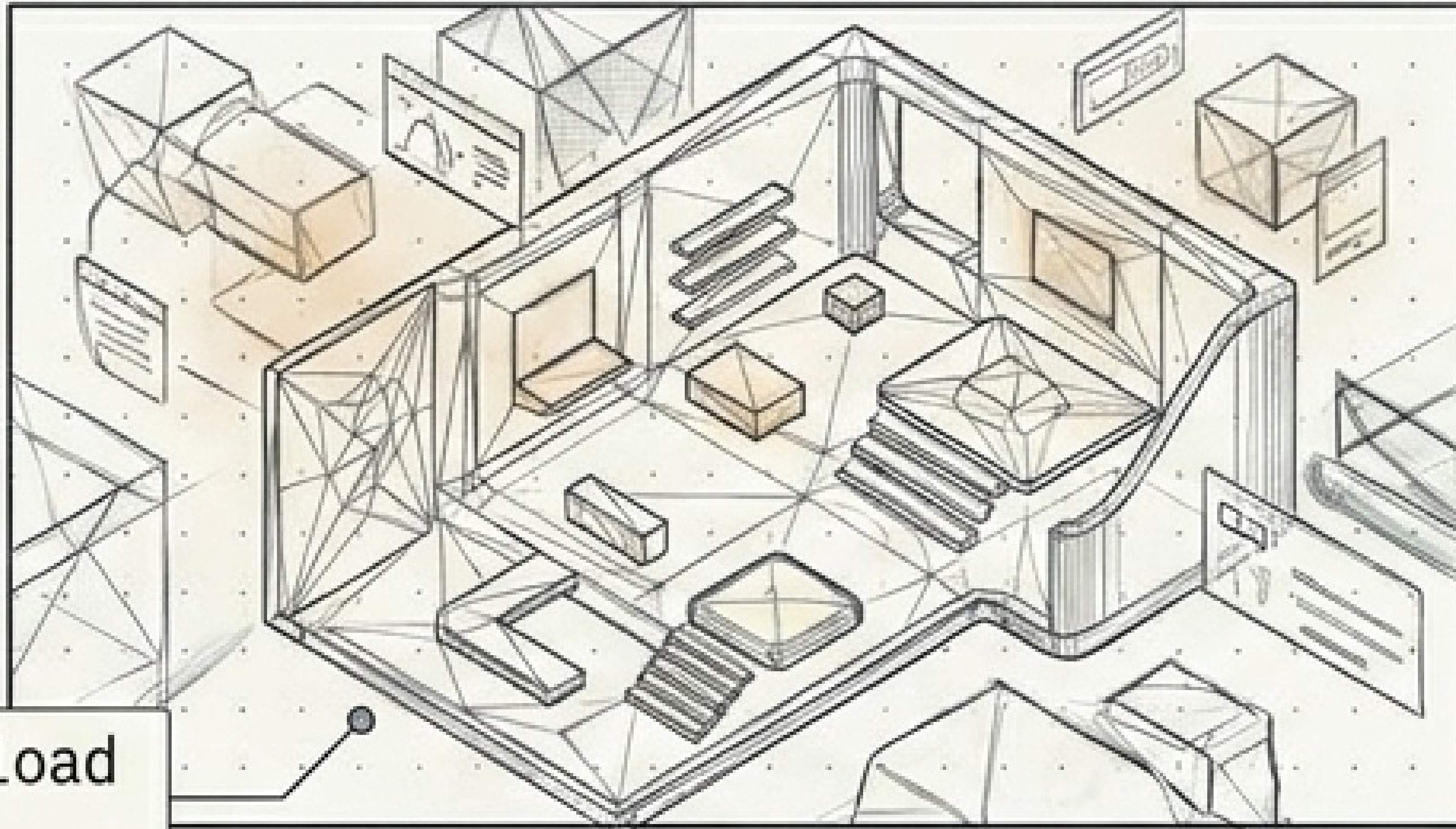


Haptic Feedback Curve

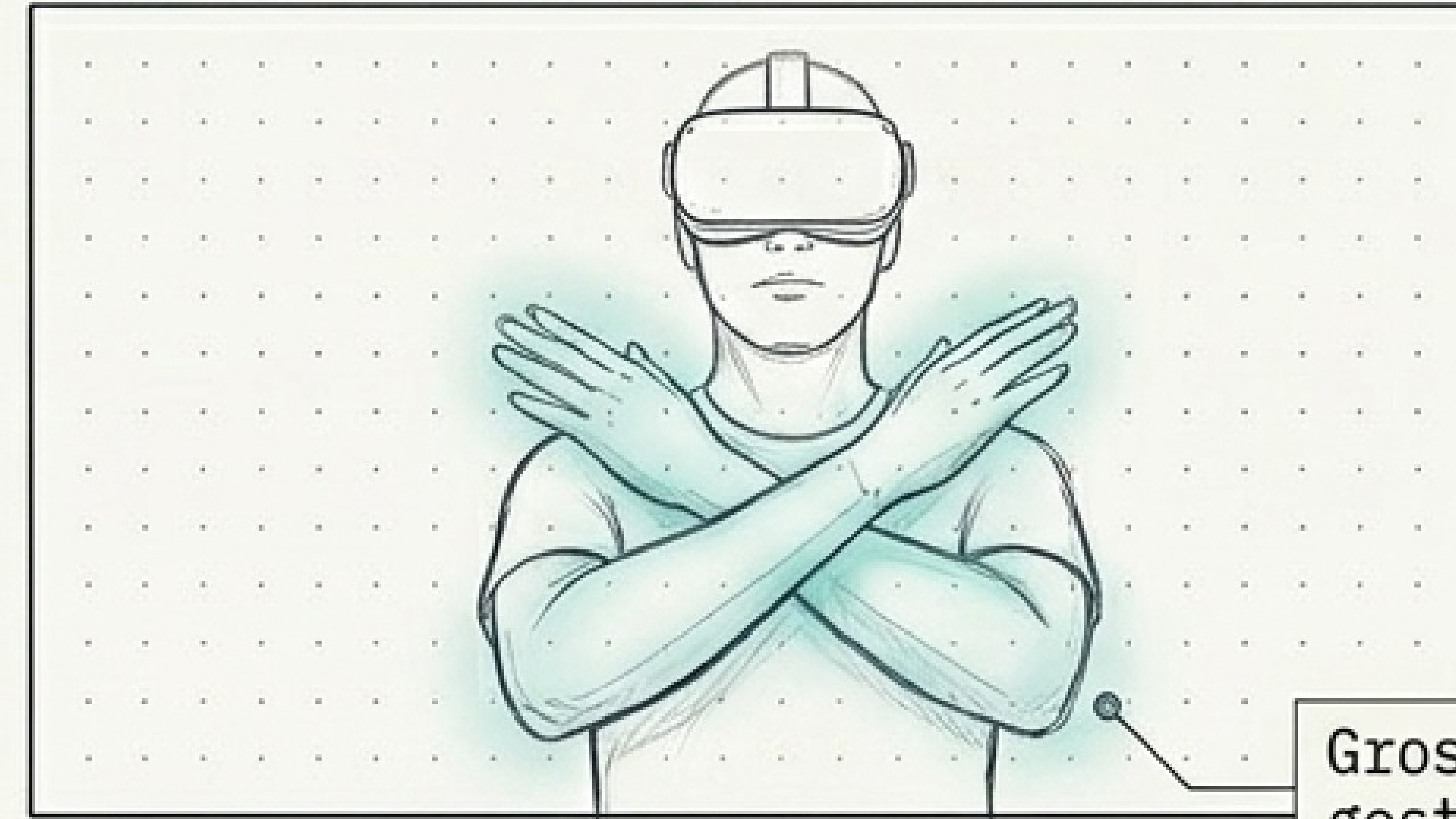


Default controller vibrations are algorithmically dampened, smoothed, and stretched out to avoid harsh tactile overstimulation during micro-interactions.

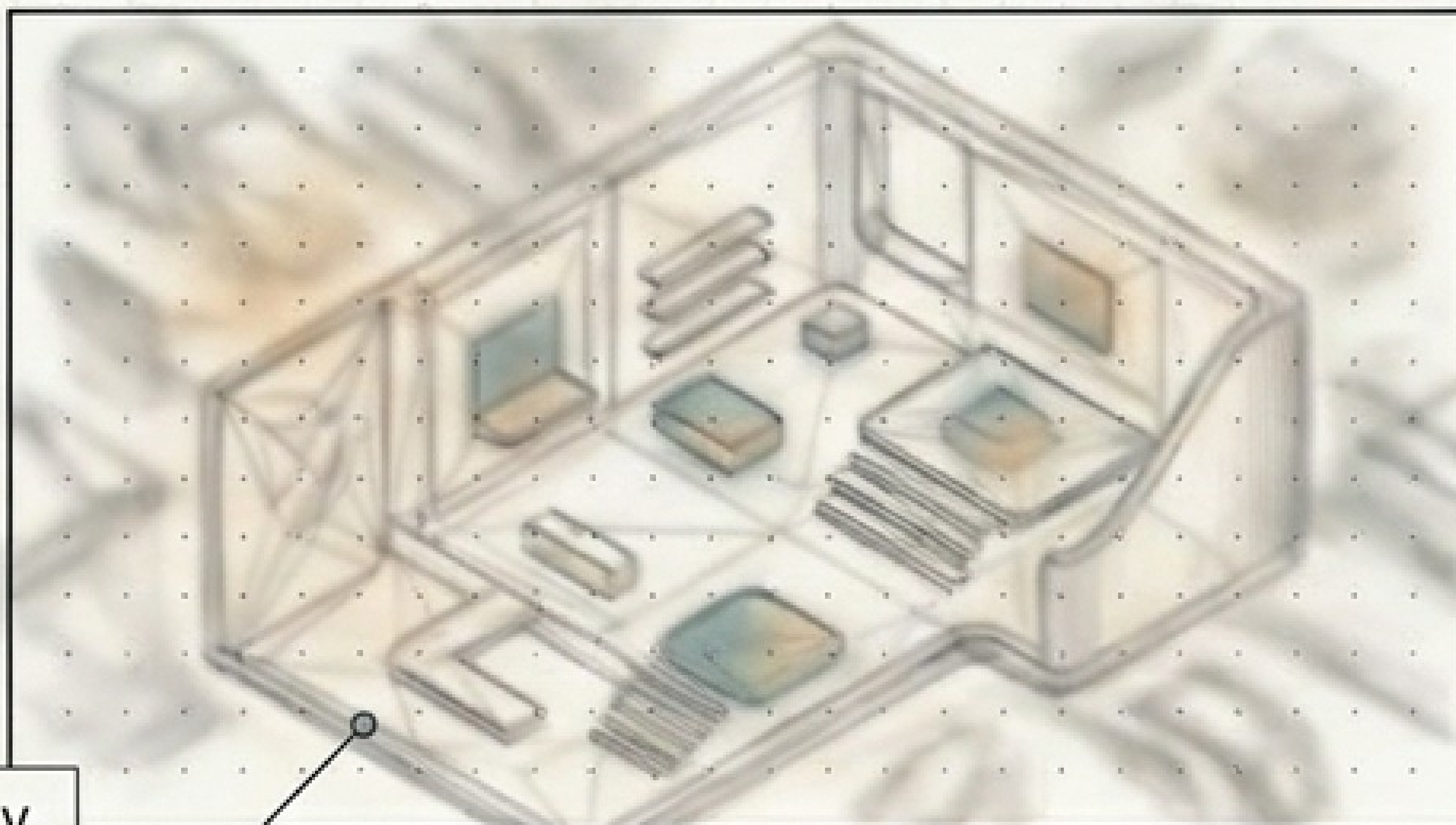
Managing Overload with Gross-Motor Triggers



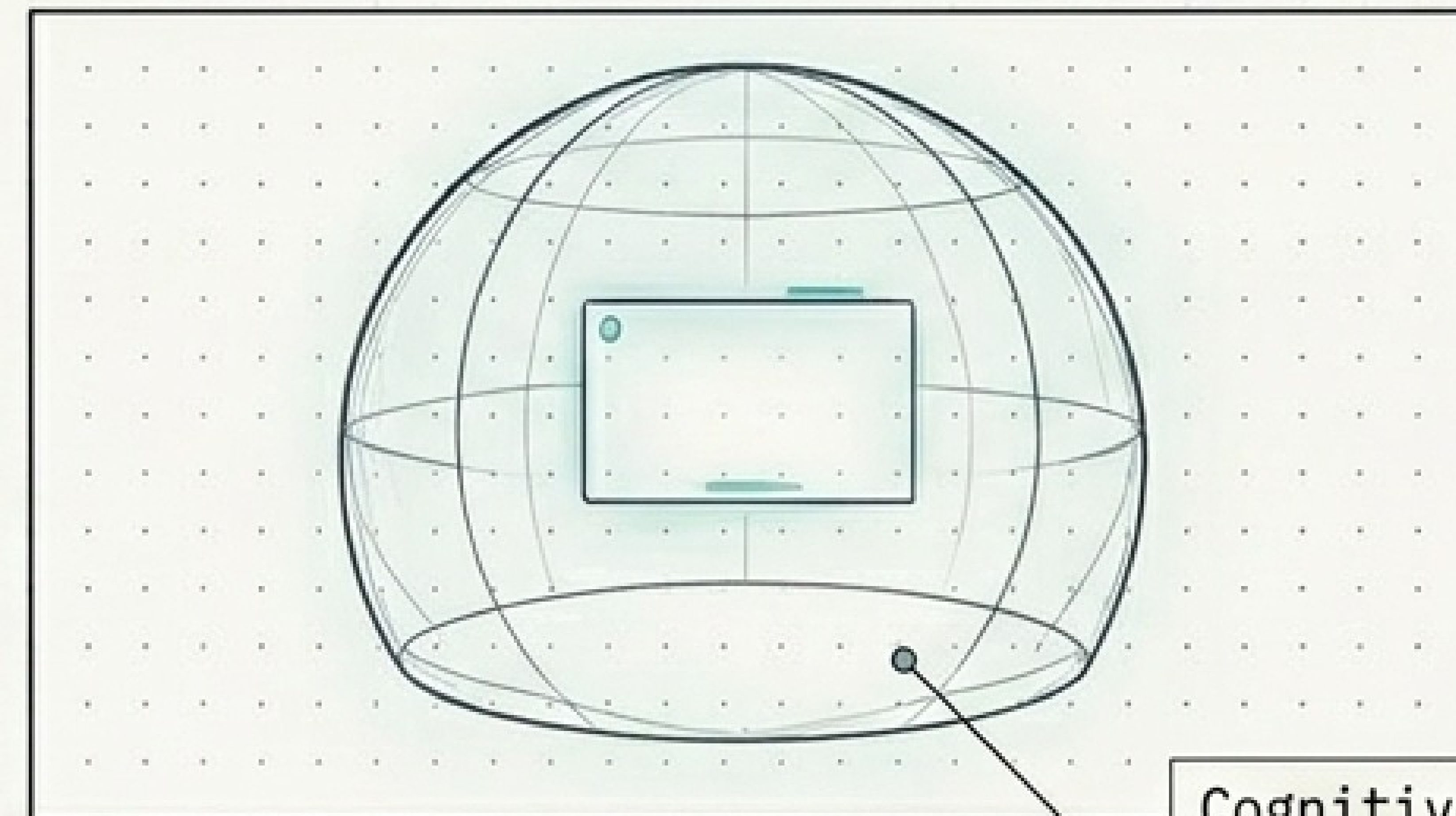
Sensory Overload Event



Gross-motor gesture trigger

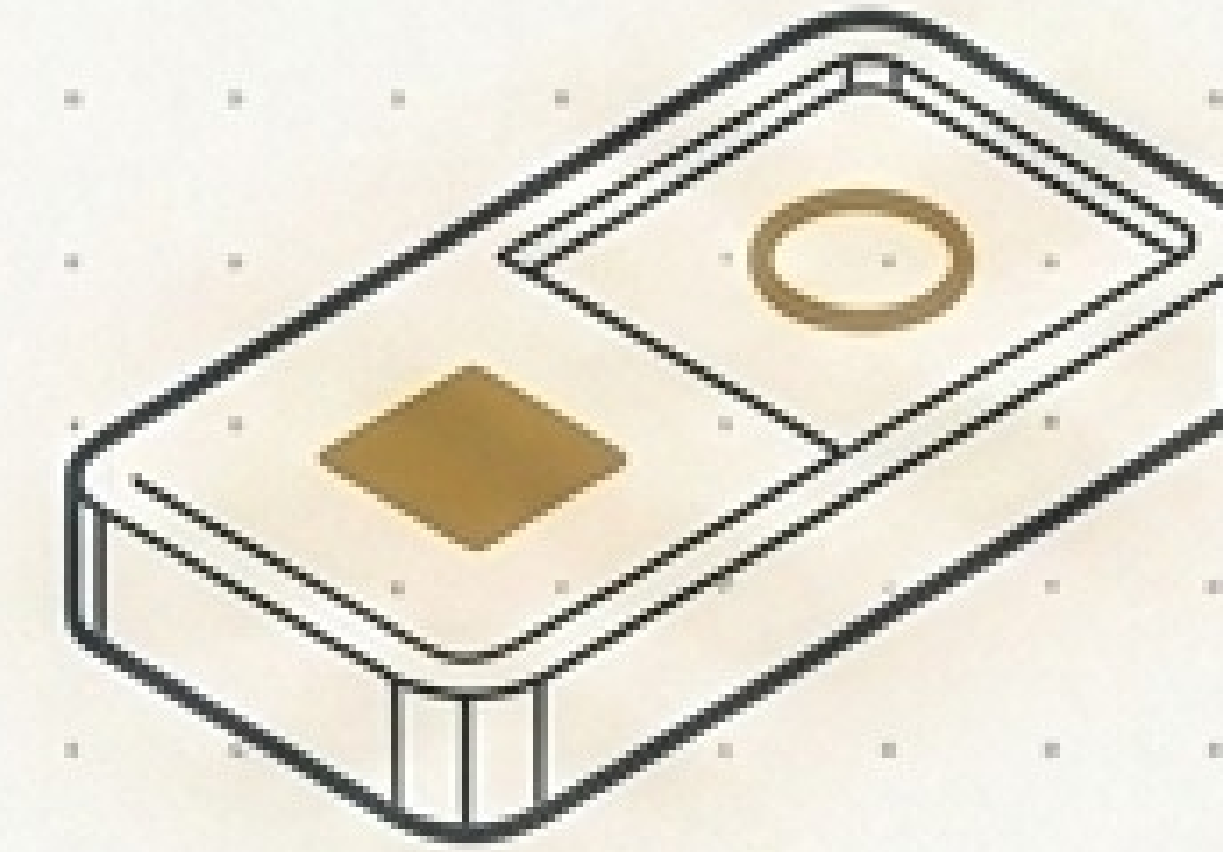


Instant sensory dampening



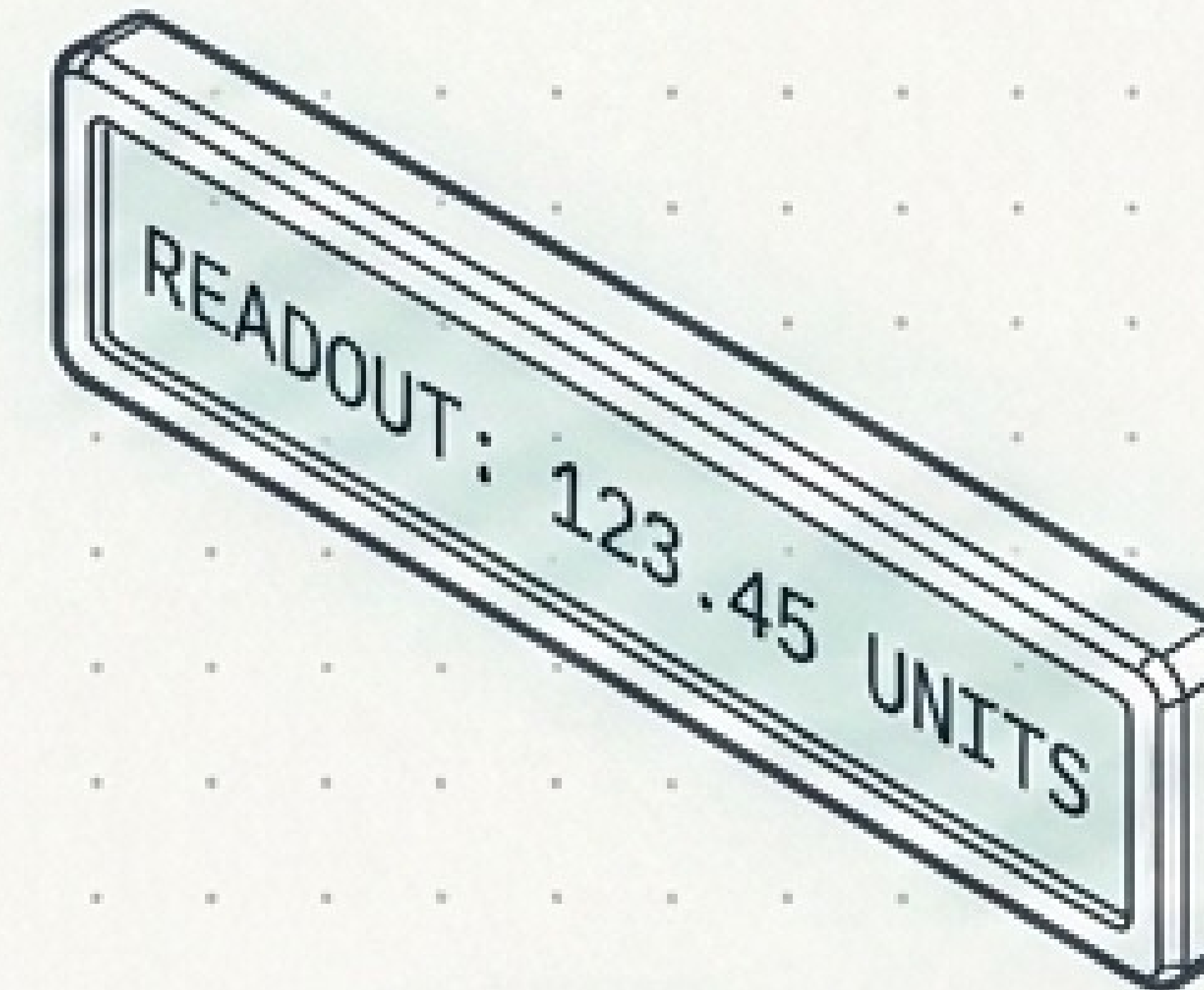
Cognitive reset protocol (Safe Room)

The Inclusive Component Library

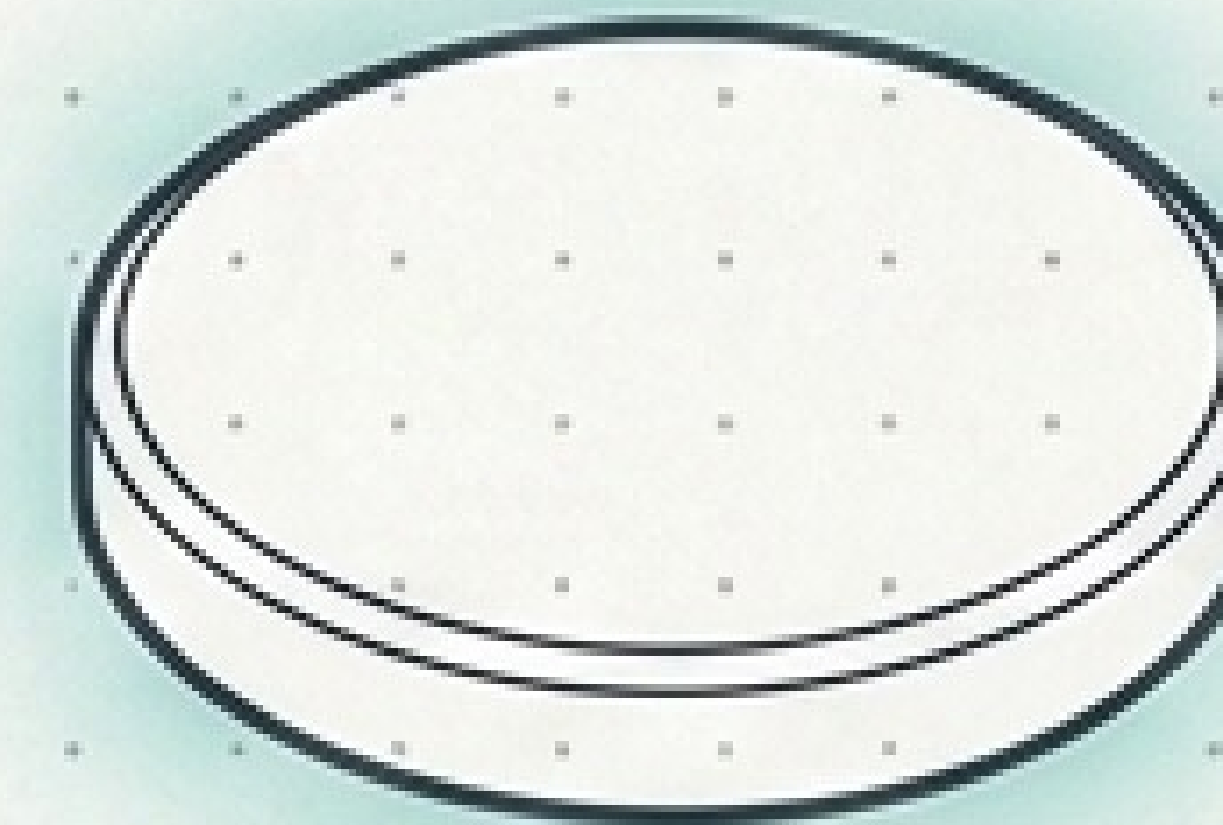


High-contrast
geometric iconography

READOUT: 123.45 UNITS

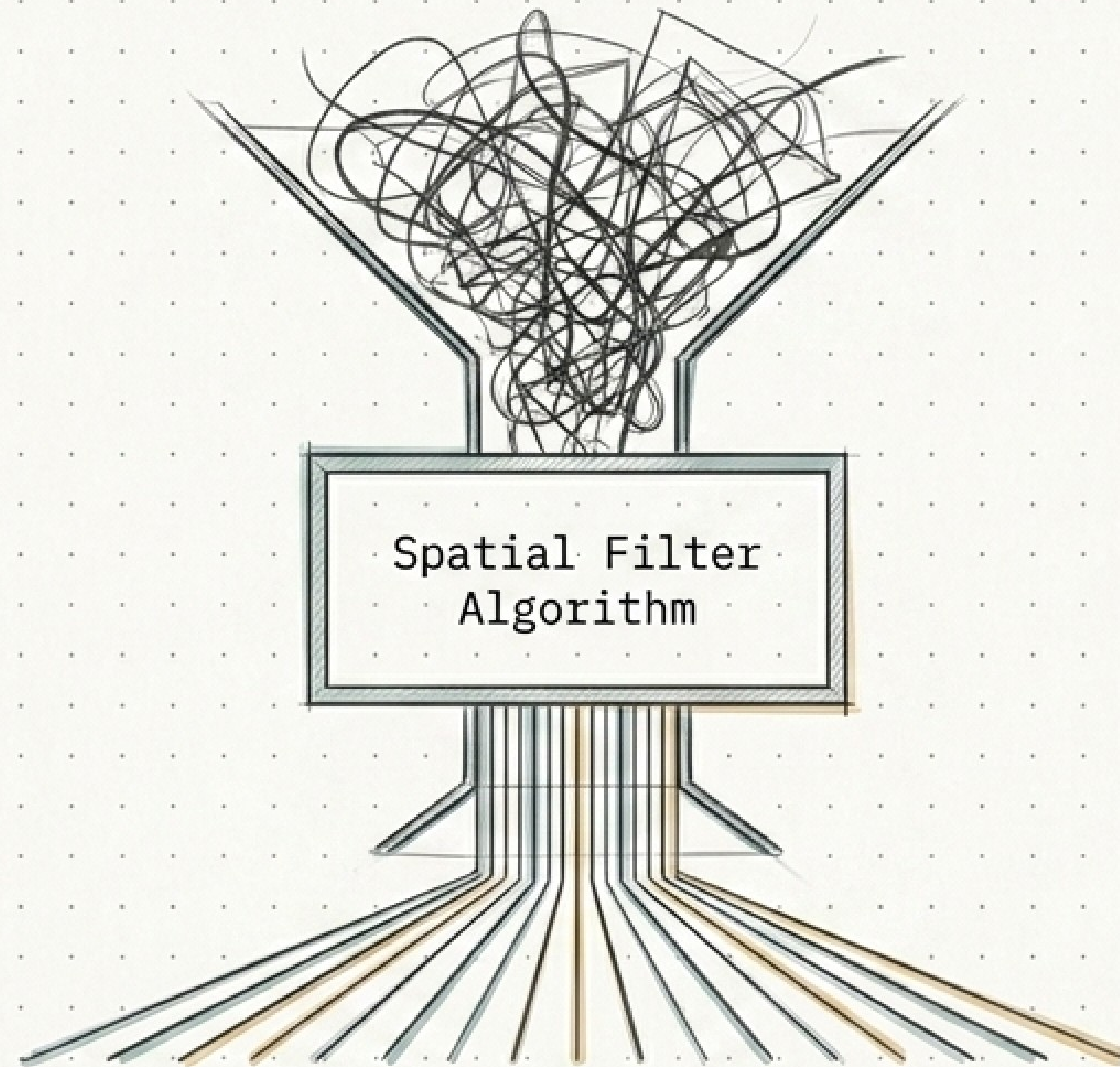


Dyslexic-friendly
monospaced typography



Total removal of
blinking or flashing
animation states

Algorithmic Dampening of Audio and Visuals



Utilizing After Effects and Unity/Oculus audio spatialization tools to isolate, dampen, and control overlapping environmental sounds before they reach user perception.

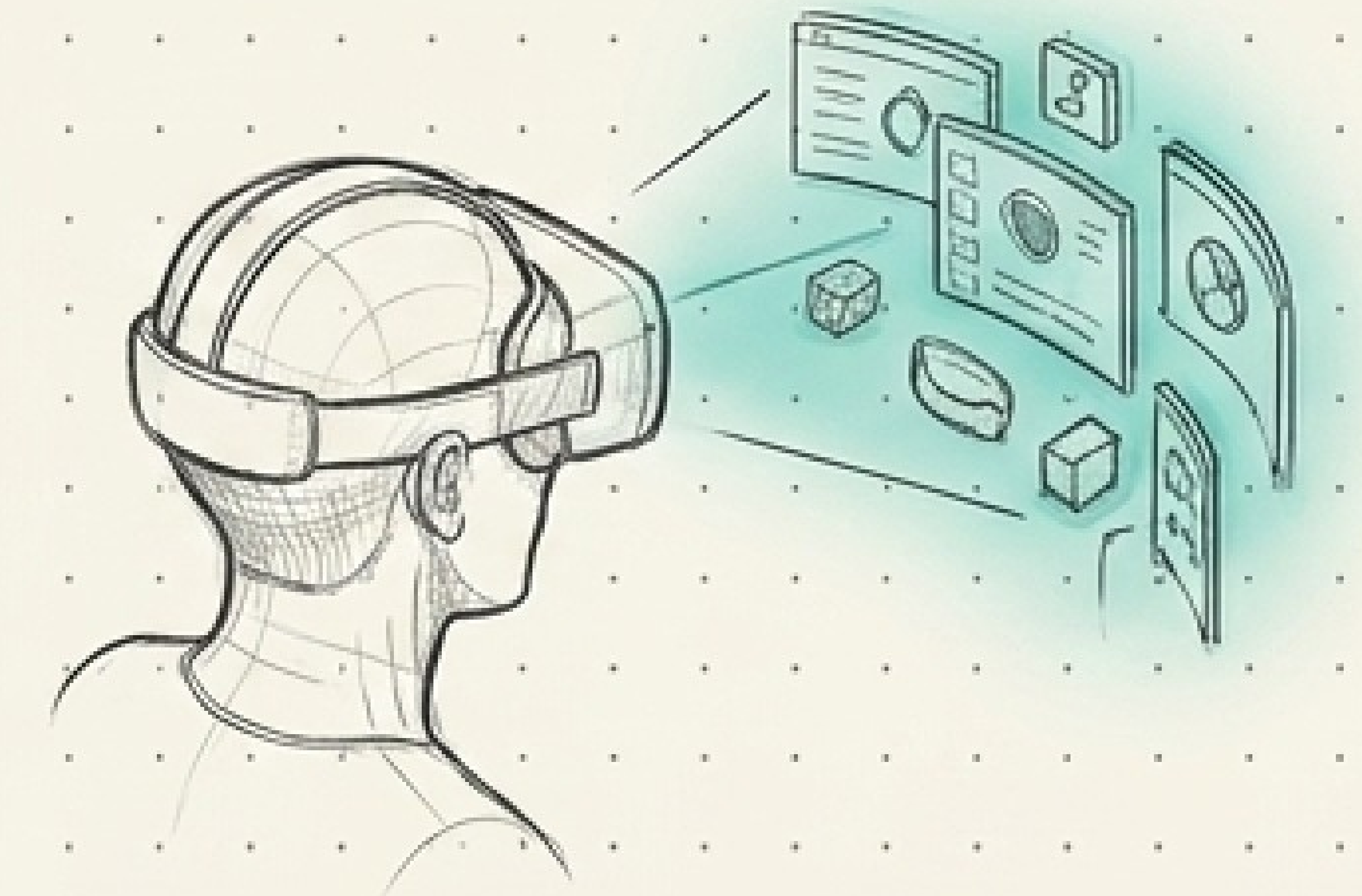
The Diagnostic Loop and Iterative Testing

Environmental Iteration



V1 Prototype

Feedback: Too much peripheral movement



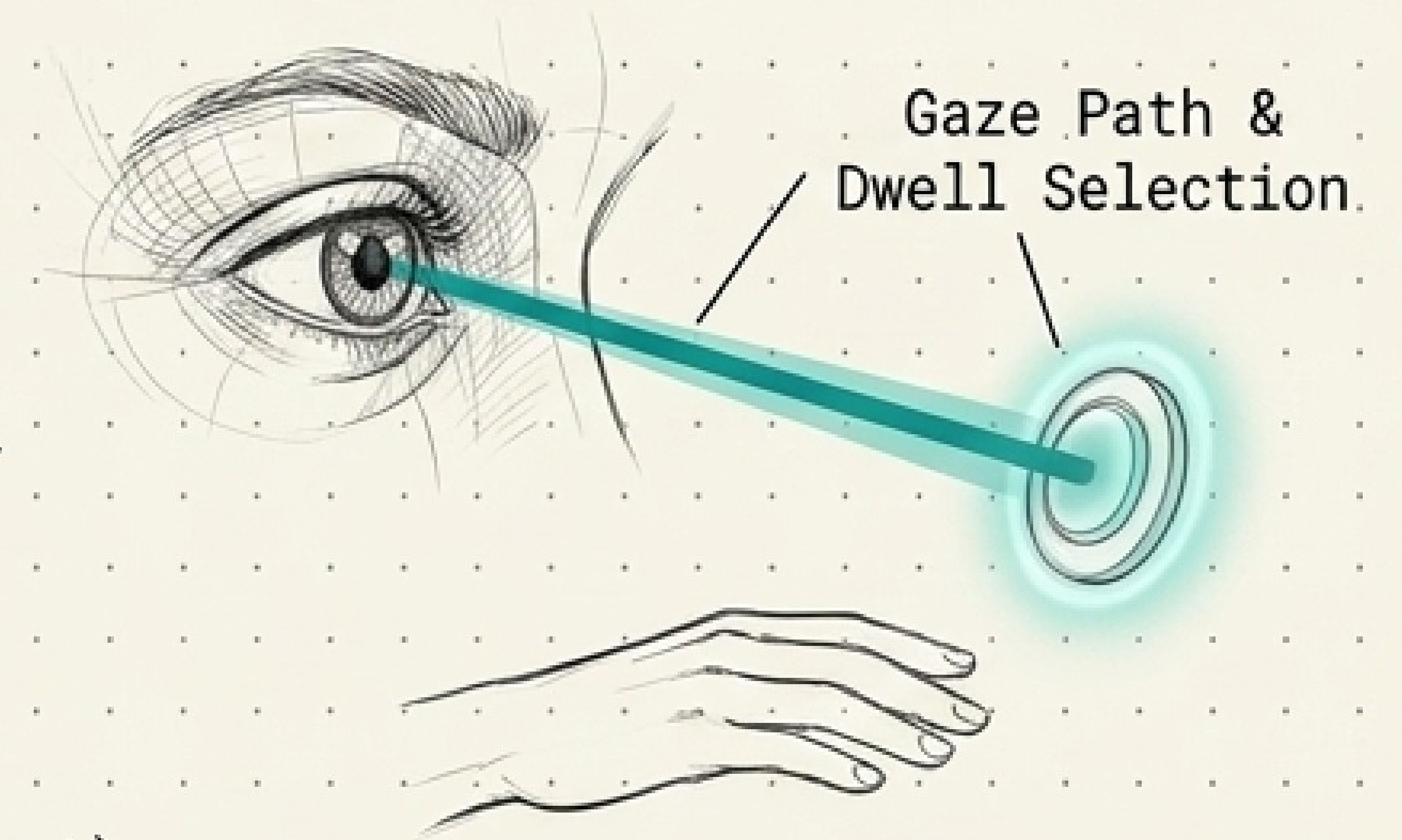
V2 Revised (Anchored Focus)

Interaction Iteration



V1 Interaction

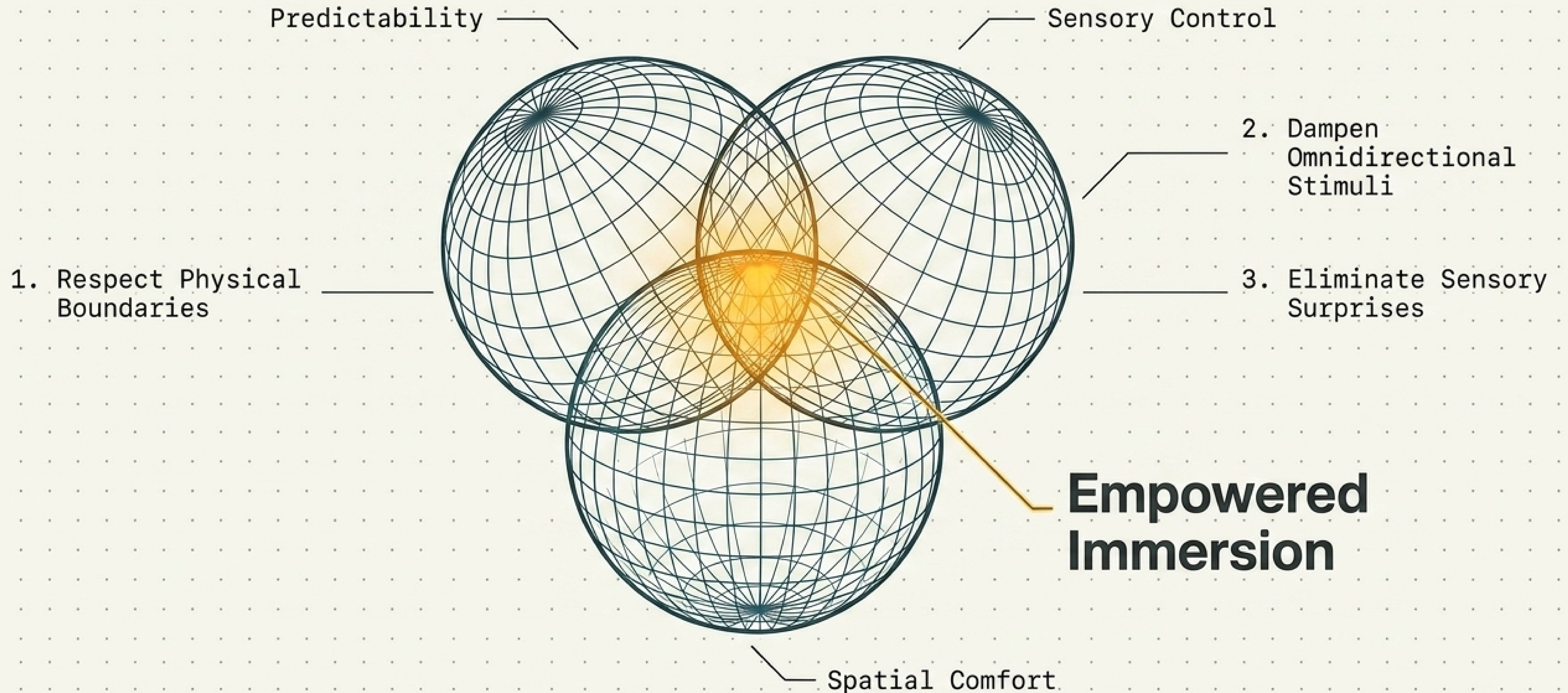
Feedback: Caused fine motor fatigue



V2 Revised (Gaze-and-Dwell)

Continuous neurodivergent user testing (NID Context) fundamentally altered spatial architecture.

The Inclusive Spatial Framework



Establishing a New Baseline for XR



True immersion shouldn't require sensory compromise. It is time to adopt inclusive architectural standards in mainstream XR development.